L2.4 Limits at Infinity *Recall: L2.3 How Derivatives Affect the Shape of a Graph*

Name ______________________________ **Solo – 9 minutes**

1. For $f(x) = x^3 - 6x^2 + 9x$, find the critical numbers and the sign of f' on each piece. Then classify each critical number by the First Derivative Test.

Write down the number you evaluated to get each sign.

2. Same function. Find every possible inflection point and the intervals of concavity, then sketch f.

3. Two questions about what happens far out.

(a) As x gets huge, what does $f(x) = x^3 - 6x^2 + 9x$ do?

(b) Now evaluate $g(x) = \dfrac{2x+1}{x-3}$ at $x = 100$ and at $x = 1000$. What is different about what g does?